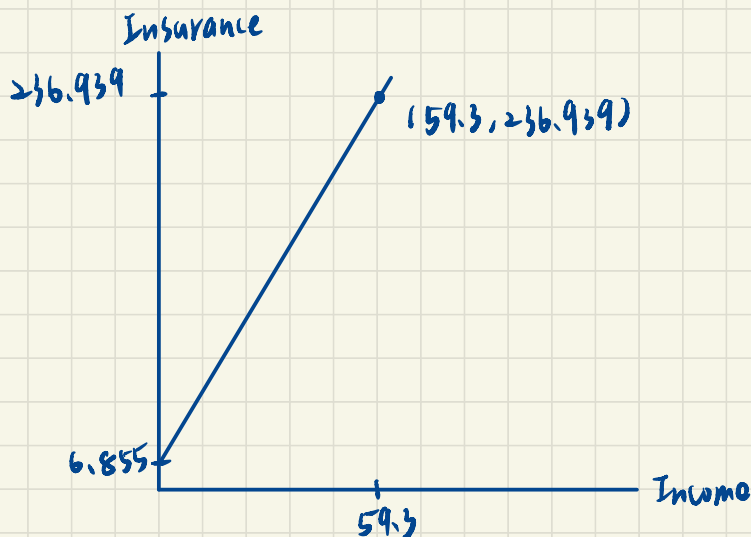


3.18

a. Sample mean of Insurance = $6.855 + 3.88 \times 59.3 = 236.939$



b. point estimate: 3.88

$$95\% \text{ CI: } [b_2 \pm t_{\alpha/2, (N-2)} \cdot se(b_2)]$$

$$= [3.88 \pm t_{0.025, 118} \cdot 0.112]$$

$$= [3.88 \pm 2.101 \cdot 0.112]$$

$$= [3.6449, 4.1153]$$

With 95% confidence that from additional \$1000 of income,

household will spend between \$3644.9 and \$4115.3 on insurance.

c.

$$\frac{\hat{y} - \bar{y}}{\hat{\sigma}_y / \sqrt{N-2}} \sim t_{\frac{1}{2}(N-2)}$$

$$\hat{y} = 6.855 + 3.88 \times 100 = 394.855$$

$$\begin{aligned} \text{Var}(\hat{y}) &= \text{Var}(b_1 + b_2 X) = \text{Var}(b_1) + 2X \text{Cov}(b_1, b_2) + X^2 \text{Var}(b_2) \\ &= 7.383^2 + 2 \cdot 100 \cdot (-0.1746) + 100^2 \cdot 0.112^2 \\ &= 30.7487 \end{aligned}$$

$$\hat{\sigma}_y = \sqrt{30.7487} = 5.5452$$

$$\begin{aligned} 99\% \text{ CI} &= [394.855 \pm 5.5452 \times t_{0.005}^{118}] \\ &= [378.8959, 410.8141] \end{aligned}$$

2.878
"

amount of insurance 99% CI = [\$ 378845.9, \$ 410814.1] #

d.

$$1. H_0: \beta_2 = 5 \quad \text{v.s.} \quad H_1: \beta_2 \neq 5$$

$$2. t = \frac{b_2 - 5}{\text{se}(b_2)} \sim t_{0.975}(118)$$

$$3. \alpha = 0.05 \quad t_{0.975}(118) = 2.101$$

$$RR = \{ t \mid |t| > t_{0.975}(118) = 2.101 \}$$

$$4. t = \frac{3.88 - 5}{0.112} = -10 \text{ ERR}$$

5. Because $t = -10 \text{ ERR}$, we reject $H_0: \beta_2 = 5$, and there have evidence to support $H_1: \beta_2 \neq 5$. The data doesn't support the claim.

e,

$$1. H_0: \beta_2 = 1 \text{ v.s } H_1: \beta_2 > 1$$

$$2. t = \frac{b_2 - 1}{\text{se}(b_2)} \sim t_{0.95(18)}$$

$$3. \alpha = 0.01 \quad t_{0.99(18)} = 2.552$$

$$RR = \{t \mid t > 2.552\}$$

$$4. t = \frac{3.88 - 1}{0.112} = 25.7143 \text{ ERR}$$

5. Because $t = 25.7143 \text{ ERR}$, we reject $H_0: \beta_2 = 1$.

and there have evidence to support $H_1: \beta_2 > 1$.

That is, as income increases the amount of insurance held increases by large amount.

3.23 $price = 93,565.9 + 0.1845 sqft^2$

Table: Regression output showing the coefficients

	Estimate	Std..Error	t.value	Pr...t...
:----- ----- ----- -----				
(Intercept)	93.565854	6.0722256	15.40882	0
I(sqft^2)	0.184519	0.0052558	35.10769	0

a. marginal effect = $2 \alpha_2 \cdot sqft$

when $sqft = 20$, marginal effect = $40 \alpha_2$

$40 \alpha_2 = 13$

1. $H_0: \alpha_2 \leq 0.325$ v.s. $H_1: \alpha_2 > 0.325$

$\alpha = 0.325$

2. $t = \frac{b_2 - 0.325}{se(b_2)} \sim t_{0.95(498)}$

3. $\alpha = 0.05$ $t_{0.95(498)} = 1.65$

$RR = |t| > 1.65$

```
> df<- df.residual(mod1)
> t<- (b2-0.325)/seb2
> tcr<- qt(1-alpha, df)
> df
[1] 498
> tcr
[1] 1.647919
> t
[1] -26.72875
```

4. $t = -26.7288$ & RR

5. Because $t = -26.7288$ doesn't fall in rejection region.

we don't reject $H_0: \alpha_2 \leq 0.325$. That is, there doesn't

have evidence to support increasing the size of 2000

square foot house by 100 square feet is greater than \$13000.

b

$$\text{marginal effect} = 2 \alpha_2 \cdot \text{sft} \cdot t$$

$$\text{when sft} = 40 \text{ marginal effect} = 80 \alpha_2$$

$$80 \alpha_2 = 13$$

$$\alpha_2 = 0.1625$$

$$1. H_0: \alpha_2 \leq 0.1625 \text{ v.s. } H_1: \alpha_2 > 0.1625$$

$$2. t = \frac{b_2 - 0.1625}{\text{se}(b_2)} \sim t_{0.95(498)}$$

$$3. \alpha = 0.05 \quad t_{0.95(498)} = 1.65$$

$$RR = \{t \mid t > 1.65\}$$

$$4. t = 4.1895 \in RR$$

```
> t <- (b2 - 0.1625) / se_b2
> t
[1] 4.189467
```

```
> p <- 1 - pt(t, df)
> p
[1] 0.0000165395
```

5. Because $t = 4.1895$ fall in reject region, and p -value less than 0.05. We reject $H_0: \alpha_2 \leq 0.1625$. That is, there have evidence to support increasing the size of 4000 square foot house by 100 square feet is greater than \$13000.

c.

$$E(\text{price} \mid \text{sfft} = 20) = 167.3735$$

$$95\% \text{ CI} = [158.0481, 176.6988]$$

```
> lowBL  
(Intercept)  
158.0481  
> upBL  
(Intercept)  
176.6988
```

In repeat sampling, about 95% interval will contain the true value of the average price of 2000 square feet house.

d. There are 3 samples with 2000 square feet, and their price are \$138000, \$169000, \$183000.

$$\text{Sample average} = 163.333$$

The sample average 163.333 falls in the 95% CI.

This indicates the quadratic regression model provides a reasonable estimation for houses of specific size,

```
> houses_20 <- collegetown[collegetown$sfft == 20, ]  
> houses_20  
  price sqft age pool fireplace close twostory  
121  138   20  8    0          0    0        0  
188  169   20  8    0          1    0        0  
262  183   20  7    0          1    0        0  
occupied  
121    0  
188    0  
262    1  
> sample_mean_20 <- mean(houses_20$price)  
> sample_mean_20  
[1] 163.3333
```

a.

--- (a) 敘述統計 ---

> # 計算 SAL1 和 APR1 的基本統計量 (包含平均數、最小、最大)

> print(summary(tuna_data[c("SAL1", "APR1")]))

SAL1		APR1	
Min.	: 1762	Min.	:0.6100
1st Qu.	: 2485	1st Qu.	:0.6900
Median	: 3136	Median	:0.8100
Mean	: 6719	Mean	:0.7825
3rd Qu.	: 8612	3rd Qu.	:0.8625
Max.	:32820	Max.	:0.9200

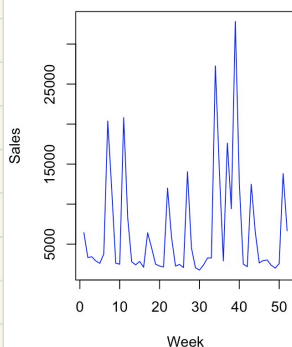
```
> sd(tuna_data$SAL1)
```

```
[1] 6915.083
```

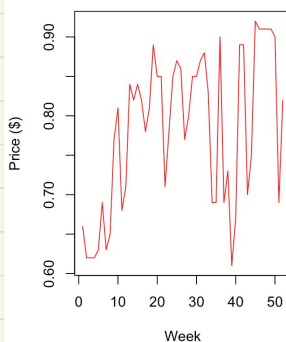
```
> sd(tuna_data$APR1)
```

```
[1] 0.09803711
```

Weekly Sales (SAL1)



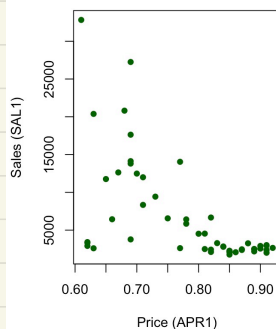
Weekly Price (APR1)



There is substantial variation in both sales and price from week to week. The sales show large volatility with sharp spikes and drops. The prices also fluctuate frequently, but less dramatically.

b.

Sales vs Price



There is negative relationship between sales and price. It is expected, because the Law of Demand, when the price increases, the quantity will decrease.

c.

```
> model<- lm(SAL1~PRICE1, data=tuna_data)
> summary(model)

Call:
lm(formula = SAL1 ~ PRICE1, data = tuna_data)

Residuals:
    Min       1Q   Median       3Q      Max
-10869.5  -1588.3  -499.3   1626.2  18607.1

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 40714.22   6196.42   6.571 2.82e-08 ***
PRICE1      -434.45    78.58   -5.528 1.17e-06 ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,    Adjusted R-squared:  0.367
F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06
```

```
> print(confint(model, level=0.95))
                2.5 %      97.5 %
(Intercept) 28268.3379 53160.0939
PRICE1      -592.2897 -276.6049
```

point estimate: $b_2 = -434.45$

95% CI = $[-592.2897, -276.6049]$

d.

90% CI = $[8624.934, 11980.87]$

```
> new_data <- data.frame(PRICE1 = 70)
> predict_ci <- predict(model, newdata=new_data, interval=
"confidence", level=0.90)
> print(predict_ci)
      fit      lwr      upr
1 10302.9 8624.934 11980.87
```

e.

point estimate = -5.059825

95% CI = $[-6.898149, -3.221501]$

$|E| > 1 \Rightarrow$ elastic

```
> mean_P <- mean(tuna_data$PRICE1)
> mean_S <- mean(tuna_data$SAL1)
> elasticity <- beta2_hat * (mean_P / mean_S)
> elasticity
PRICE1
-5.059825
> se_elasticity <- se_beta2 * (mean_P / mean_S)
> df <- df.residual(model)
> tcr <- qt(0.975, df)
> lower_E <- elasticity - tcr * se_elasticity
> lower_E
PRICE1
-6.898149
> upper_E <- elasticity + tcr * se_elasticity
> upper_E
PRICE1
-3.221501
```

It makes economic common sense. Because tuna has many substitutes, customer are more sensitive to quantity

When the price changes.

f.

1. $H_0: \epsilon = -3$ v.s $H_1: \epsilon \neq -3$

2.
$$t = \frac{\hat{\epsilon} - (-3)}{se(\hat{\epsilon})} \sim t_{0.95(150)}$$

3. $\alpha = 0.1 \quad t_{0.95(150)} = 1.6759$

$$RR = \{t \mid |t| > 1.6759\}$$

```
> alpha<- 0.1
> elasticity <- beta2_hat * (mean_P / mean_S)
> se_elasticity <- se_beta2 * (mean_P / mean_S)
> df<- df.residual(model)
> t<- (elasticity-(-3))/se_elasticity
> t
      PRICE1
-2.250572
> tcr<- qt(1-(alpha/2), df)
> tcr
[1] 1.675905
> p <- 2*(1-pt(abs(t), df))
> p
      PRICE1
0.02884204
```

4. $t = -2.25 < t_{RR}$

5. Because $t = -2.25$ fall in reject region.

and $p\text{-value} = 0.0288$ less than 0.1 .

We reject $H_0: \epsilon = -3$.

There have evidence to support $H_1: \epsilon \neq -3$. That is the price elasticity is significant different from -3 .